Circular Measure & Trigonometric Ratios

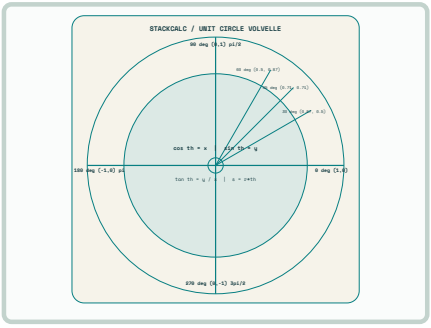
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.HSF.TF.A.1

Student Name: _____ Date: _____ **Model: SC-32** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-32 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Unit Circle Volvelle

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Rotate outer acrylic volvelle rotor to 30°; read exact coordinate (√3/2, 1/2) in window.

Sketch Model on Paper:
Draw unit circle showing angle theta, reference triangle, and arc length s = rtheta.

3. Step-by-Step LCD Stack Trace (SC-32)

MODE: 123 Base-10 Arithmetic

1. 45 X: 45 Key input operand (n
2. COS X: 0.7071 Evaluate monadic COS

EXPECTED LCD X: 0.7071 (√2/2)

1 **Find sin(30°) on SC-32 in DEG mode.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-32

2 **Find cos(60°) on SC-32 in DEG mode.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-32

3 **Find tan(45°) on SC-32 in DEG mode.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-32

4 **Convert 180° to radians: [180] [▣RAD].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-32

Station Diagnostic & Solution Key: Chapter 7

Circular Measure & Trigonometric Ratios • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Unit Circle Trigonometry Volvelle

CAD Blueprint: unit_circle_volvelle

Target Device: SC-32 Scientific

CAD / STL Source File:

Hardware/models/stl/accessories/volvelle.stl

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Dual Precision Rotary Dial in Dark Blue & White (PLA/PETG)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Find sin(30°) on SC-32 in DEG mode.** Standard: CCSS.MATH.HSF.TF.A.1	0.5000	[30] [SIN]
2	**Find cos(60°) on SC-32 in DEG mode.** Standard: CCSS.MATH.HSF.TF.A.1	0.5000	[60] [COS]
3	**Find tan(45°) on SC-32 in DEG mode.** Standard: CCSS.MATH.HSF.TF.A.1	1.0000	[45] [TAN]
4	**Convert 180° to radians: [180] [⊠RAD].** Standard: CCSS.MATH.HSF.TF.A.1	3.1416 (π rad)	[180] [⊠RAD]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Always check whether calculator angle mode is set to DEG or RAD before computing.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-32, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.